

threshold value. It will keep adding the replicas till the maximum pods limit is reached, as defined in the HPA configuration.

The entire walkthrough for HPA is as follows.

We have a service called *php-apache* and deployment called *php-apache*, which has a resource limit of 200m CPU. We will be setting up a 50 per cent threshold limit in our HPA configuration, as follows:

```
apiVersion: v1
kind: Service
metadata:
  creationTimestamp: null
  name: php-apache
spec:
  ports:
    - port: 80
      protocol: TCP
      targetPort: 80
  selector:
    run: php-apache
status:
  loadBalancer: {}
---
---
apiVersion: apps/v1
kind: Deployment
metadata:
  creationTimestamp: null
  labels:
    run: php-apache
  name: php-apache
spec:
  replicas: 1
  selector:
    matchLabels:
      run: php-apache
  strategy: {}
```

```
template:
  metadata:
    creationTimestamp: null
    labels:
      run: php-apache
  spec:
    containers:
      - image: k8s.gcr.io/php-apache
        name: php-apache
        ports:
          - containerPort: 80
        resources:
          requests:
            cpu: 200m
status: {}
```

To create the service and deployment together, we can use:

```
kubectl create -f hpaapp.yml
```

Given below is the HPA example, which is targeting *php-apache* deployment with 50 per cent threshold value. The moment this deployment reaches 50 per cent of the assigned resource (200m in our case), it will trigger an additional pod.

```
apiVersion: autoscaling/v1
kind: HorizontalPodAutoscaler
metadata:
  creationTimestamp: null
  name: php-apache
spec:
  maxReplicas: 10
  minReplicas: 1
```

```
scaleTargetRef:
  apiVersion: apps/v1
  kind: Deployment
  name: php-apache
  targetCPUUtilizationPercentage: 50
status:
  currentReplicas: 0
  desiredReplicas: 0

kubectl create -f hpa.yml
kubectl get hpa
```

To test this, we can use a temporary pod called *load-generator* and run a *while* loop for a few minutes. Then we can exit the pod and check whether it got autoscaled.

 **Note:** If the load decreases, and the number of pods is above the configured minimum, the HPA will scale down. This value can be configured using the `—horizontal-pod-autoscaler-downscale-stabilization` flag, which defaults to 5 minutes.

Run the command given below to see if autoscaling is working as desired:

```
kubectl run -i --tty load-generator
--image=busybox /bin/sh
while true; do wget -q -O- http://php-apache.default.svc.cluster.local; done
```

You can try out different threshold limits and configurations as per your requirement. **END** 

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